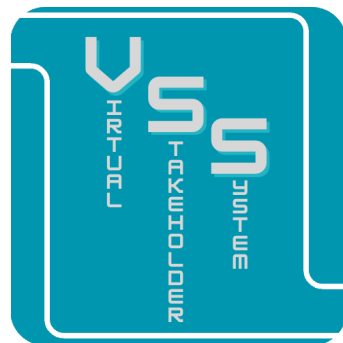




Virtual Stakeholder System

Group #28

Project Deliverable #1

September 29th, 2024

1. Vision:

The Virtual Stakeholder System (VSS) will aim to replicate the role of real-world stakeholders, who may be missing or hard to reach in practice. The system intends to allow users to interact with simulated stakeholders that take on the experience and knowledge of real ones, providing relevant information that helps contextualize requirements. Just like a requirements engineer, the VSS will be able to create essential artifacts such as non-functional requirements, personas, user stories and use case scenarios for the users of the system, so they can better streamline their development processes. In addition, the VSS will be able to test for the completion status of the given requirements, as well as the coherence of generated requirements, which allows users to receive vital feedback on their project.

2. Preliminary Requirements:

ID	Requirement	Guidelines/ Additional Notes	Source	Priority (High, Medium, Low)	Type
1	The system will allow users to specify goals for their project.	- On user request	Lecture 3 slides	High	Functional
2	The system will automatically save stakeholder instances between sessions.		Brainstorming Session	High	Functional
3	The system will allow users to save up to 15 stakeholder instances.		Brainstorming Session	Medium	Non-functional
4	The LLM will support both English and French for all inputs and all outputs.		Brainstorming Session	High	Non-functional
5	The system will allow users to ask the virtual stakeholder questions.	- By training a large language model	Project description Document - Objective 2	High	Functional
6	The system will provide answers to users' questions.	- Acting as the virtual stakeholder	Project description Document - Objective 2	High	Functional
7	The system will allow users to create new stakeholder personas.	- By training a large language model	Project description Document - Objective 3	High	Functional
8	The system shall allow users to specify characteristics of created personas.	- On user request	Project description document - Objective 3	Medium	Functional
9	The system will automatically produce persona artifacts.	- On user request	Project description Document - Objective 3	High	Functional

10	The system will automatically update persona artifacts throughout the development process.	- Through data gathered by the LLM	Project description Document - Objective 3	High	Functional
11	The system will automatically produce user stories.	- On user request	Project description Document - Objective 3	High	Functional
12	The system will automatically produce non-functional requirements.	- On user request	Project description Document - Objective 3	High	Functional
13	The system will automatically produce use case scenarios.	- On user request	Project description Document - Objective 3	High	Functional
14	The system should produce requirement deadlines based on the project outline input by the user.	- Deadlines should include time (24 hours), day, month and year	Brainstorming session	Medium	Functional
15	The system must be able to validate certain requirements as specified by the user.	- On user request	Product description Document - Objective 4	High	Functional
16	The system shall automatically detect duplicate and highly similar requirements and notify users of them.	- By training a large language model on textual analysis and contextual data	Product description Document - Objective 4	High	Functional
17	The system shall automatically detect requirements which are not coherent with each other and notify users of them.	- By training a large language model on textual analysis and contextual data	Product description Document - Objective 4	Medium	Functional
18	The system will incorporate human feedback for validation of individual or collective requirements.	- By allowing users to leave feedback on requirements	Requirements Validation & challenges with inconsistent information	Medium	Functional
19	The system shall ensure all information on requirements and artifacts are encrypted during storage and transmission of data.	- Using industry standard security protocols	Product description Document - Objective 1	High	Non-functional

20	The large language model will not be trained on any personal user data to keep the stakeholder's identities safe.	- Data includes: full name, address, email address and phone number	Brainstorming session	Low	Non-functional
21	The system's virtual stakeholder must generate a response to a user's question in no more than 15 seconds.	- On user request	Industry Standards	High	Non-functional
22	The system shall produce requested artifacts in no more than 15 seconds.	- On user request	Project description document - Objective 3	High	Non-functional
23	The system shall have 99.9% uptime.		Industry Standards	High	Non-functional
24	The system will maintain an error rate of no more than 0.1% for user requests.		Industry Standards	High	Non-functional
26	The system will be tested regularly to mitigate data changes and update responses based on newly identified benchmarks		Anticipated Challenges with Inconsistencies	High	Non-functional
27	The LLM will be trained with user responses based on different gender types to produce unbiased personas.		Social,Cultural or Gender Biases	Medium	Non-functional
28	The LLM will be trained with feedback from culturally diverse human reviewers to provide a balanced response aligning with the cultural values and ethics.		Social,Cultural or Gender Biases	High	Non-functional
29	The system will be trained with data from at least 50 specialized unfamiliar domains to generate responses as specialized stakeholders.		Gather Information on Unfamiliar Domains	High	Non Functional

3. Discussion:

First we all met up in person for a brainstorming session. We started by discussing our overall vision for the project and the direction we wanted to go in. From here, we compiled a list of requirements obtained from a brainstorming session. We then revised the list of requirements and selected those which we agreed would most benefit the system. One of the strategies we applied to filter our requirements was to ask another member of our team if they thought the requirement was viable. If it made sense to the team we added it to the table. We then reviewed the project description document and added requirements based on each specification. Overall, our collective strategies proved both very effective, and very efficient. They allowed us to work through the deliverable with little to no issues as a group.

4. Team Description and Roles:

Who they are?	Tasks	Percent
Anna	- Document Creation, Formatting and forwarding - Requirements - Discussion	20%
Alex	- Requirements - Logo and Page Stylization - Formatting	20%
Caleb	- Requirements - Discussion - Grammar Check	20%
Owen	- Vision - Requirements - Formating	20%
Mukhveer	- (Quality and Performance) Non Functional Requirements - Grammar Check - Formating	20%